MILITARY SPECIFICATION

# HOUSING, TIMING AND FUZING; IGNITOR ASSEMBLY AND SUSTAINER CHARGE ASSEMBLY FOR

This specification is approved for use by the U.S. Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 This specification contains requirements not normally covered by the drawings and provides quality assurance provisions for the fabrication of parts, assembly and packing for Igniter Assembly and Sustainer Charge Assembly.

## 2. APPLICABLE DOCUMENTS

2.1 Issues of documents. - The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

| | | |
|---|---|---|
| MIL-A-148 | - | Aluminum foil |
| MIL-P-223 | - | Powder, Black |
| MIL-A-48078 | - | Ammunition, Standard Quality Assurance Provisions, General Specification For |

FSC: 1345

**Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.**

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts
MIL-STD-650 - Explosive: Sampling, Inspection and Testing

DRAWINGS (see 6.4)

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

PRODUCT AND PACKING DRAWINGS

8796522 - Marking Diagram and Sealing for Wood Packing Boxes
9298549 - Sustainer Charge Assembly
9298551 - Igniter Assembly

INSPECTION EQUIPMENT DRAWINGS

9298424 - Closed Bomb Fixture

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

TITLE 49 - Transportation, Parts 100-199

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402, Orders should specify, "49 CFR 100-199 (latest revision)").

3. REQUIREMENTS

3.1 Material. - Materials shall be in accordance with the applicable drawings and specifications.



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3.2 Assembly. - The assemblies shall comply with all requirements specified on Drawing (Dwg) 9298549 and 9298551 all associated drawings, and with all requirements specified in applicable specifications.

3.3 Moisture content of lead styphnate (Ignitor Assembly).- The moisture content of the lead styphnate at the loading station at the time of loading shall not exceed 0.30 percent.

3.4 Output pressure/time test of ignitor assembly and sustainer charge assembly and pairing of serial lots. -

3.4.1 Output pressure/time test. - When the igniter assembly and sustainer charge assembly are assembled in a closed bomb with internal volume of 0.122 cubic inches they shall produce a pressure/time trace which has a peak pressure of 500 psig (+70, -60 psig) at 165 to 220 milliseconds from the time of ignition when at a temperature of 70 ± 5°F. In addition the assemblies shall produce a peak pressure of 400 psig minimum at 165 to 220 milliseconds from the time of ignition when the assemblies are at a temperature of -25 ± 5°F.

3.4.2 Pairing of ignitor assembly and sustainer charge assembly serial lots. - After successful completion of the output pressure/time test and all other quality conformance inspections, each igniter assembly serial lot and sustainer charge assembly serial lot tested together shall be paired for use in the housing, timing, and fuzing and shall contain equal quantities. All applicable shipping documents shall specify those serial lots paired for use together.

3.5 First Article Inspection. - This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.6 Workmanship. - All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust, and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous; however, a few light scratches not exposing base material may be permitted. All required marking and stamping shall be neat and sharply defined.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions. - Unless otherwise specified herein in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspections. - The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First Article Inspection. -

4.3.1 Submission. - The contractor shall submit a first article as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

| Part Description | Drawings | Quantity |
|---|---|---|
| Header | 9298602 | 25 |
| Cup, Igniter | 9298552 | 25 |
| Cup, Sustainer | 9298550 | 25 |
| Sustainer Charge Assembly | 9298549 | 70 |
| Igniter Assembly | 9298551 | 250 |

4.3.2 Inspections to be Performed. - See MIL-A-48078 and Table 1 herein.

4.3.3 Rejection, - See MIL-A-48078.



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TABLE 1 - FIRST ARTICLE INSPECTION
CLASSIFICATION OF DEFECTS & TESTS

| PARAGRAPH | TITLE | | SHEET 1 OF 2 | | DRAWING NUMBER |
|---|---|---|---|---|---|
| | Ignitor Assembly and Sustainer Charge Assembly and Components | | | | See Below |
| | | | | | NEXT HIGHER ASSEMBLY |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| | Cup, Ignitor | | | | |
| | (Dwg. 9298552) | | | | |
| | Examination for Defects | 25 | | | 4.4.2.1 |
| | Header | | | | |
| | (Dwg. 9298602) | | | | |
| | Pull test of electrodes | 25 | | | 4.5.3 |
| | Ignitor Assembly | | | | |
| | (Dwg. 9298551) | | | | |
| | Bridgewire Resistance | 150 | | | 4.5.4 (Note 1) |
| | Insulation Resistance | 150 | | | 4.5.5 (Note 1) |
| | Insulation Resistance, thermal shock, vibration and function | 25 | | | 4.5.6 |
| | All fire Voltage (ambient) | 125 | | | 4.5.7 |
| Note 1: | The insulation and bridgewire resistance tests shall be performed on all ignitor assemblies prior to performing all other listed ignitor assembly tests. | | | | |

NOTES:





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TABLE 1 - FIRST ARTICLE INSPECTION

**CLASSIFICATION OF DEFECTS & TESTS** MIL-H-48384(AR)

| PARAGRAPH | TITLE | | SHEET 2 OF 2 | | DRAWING NUMBER |
|---|---|---|---|---|---|
| | Ignitor Assembly and Sustainer Charge Assembly and Components | | | | See Below |
| | | | | | NEXT HIGHER ASSEMBLY |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| | Cup, Sustainer | | | | |
| | (Dwg. 9298550) | | | | |
| | Examination for Defects | 25 | | | 4.4.2.7 |
| | Sustainer Charge Assembly | | | | |
| | (Dwg. 9298549) | | | | |
| | Examination for Defects | 25 | | | 4.4.2.8 |
| | Ignitor Assembly and Sustainer Charge Assembly | | | | |
| | Output Pressure/Time Test | | | | |
| | Ambient (70°F ± 5°F) | 20 | | | 4.5.8 |
| | Cold (-25°F ± 5°F) | 20 | | | 4.5.8 |

NOTES:

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4.4 Quality Conformance Inspection. -

4.4.1 Inspection Lot Formation. - Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of Igniter Assemblies shall contain:

a. Parts from one lot interfix number from one manufacturer.
b. Milled lead styphnate (loose charge) from one batch or blending of batches which results in homogeneity of all milled lead styphnate used in loading the entire igniter assembly lot.
c. Lead styphnate with lacquer (spot charge) from one batch or blending of batches which results in homogeneity of all lead styphnate with lacquer used in loading the entire igniter assembly lot.

Inspection lots of Sustainer Charge Assemblies shall contain:

a. Sustainer cups of one interfix lot number from one manufacturer.
b. Black powder from one batch or blending of batches which results in homogeneity of black powder used in loading the entire sustainer charge assembly lot.

NOTE: Igniter assemblies and sustainer charge assemblies shall be from serial lots which are to be paired in final item production, No mixing of lots is permitted.

4.4.2 Examination. - See MIL-A-48078.

a. Sampling Plans. - Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.



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QUALITY CONFORMANCE INSPECTION

# CLASSIFICATION OF DEFECTS & TESTS

MIL-H-48384(AR)

| PARAGRAPH | TITLE | | SHEET 1 OF 1 | | DRAWING NUMBER |
|---|---|---|---|---|---|
| 4.4.2.1 | Cup, Ignitor | | | | 9298552 |
| | | | | | NEXT HIGHER ASSEMBLY |
| | | | | | 9298549 |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE / INSPECTION METHOD |
| Critical | None defined | | | | |
| Major | | | | | |
| 101 | Thickness through closed end | | 0.40% | 3.2 | Gage |
| 102 | Surface finish on outside of closed end | | 0.40% | 3.2 | Gage/Witness Blocks |
| 103 | Material thickness | | 0.40% | 3.2 | Gage |
| 104 | Cup leaks with specified air pressure applied | | 0.40% | 3.2 | Gage |
| Minor | | | | | |
| 201 | Evidence of poor workmanship | | 0.65% | 3.6 | Visual |

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| PARAGRAPH | TITLE | SHEET | DRAWING NUMBER |
|---|---|---|---|
| 4.4.2.2 | Header, bridged | 1 OF 1 | 9298484 |
| | | | NEXT HIGHER ASSEMBLY: 9298500 |

| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
|---|---|---|---|---|---|
| Critical | None defined | | | | |
| Major | | | | | |
| 101 | Bridgewire missing | | 0.40% | 3.2 | Visual |
| 102 | Bridgewire weld missing or inadequate | | 0.40% | 3.2 | Visual/Manual |
| Minor | | | | | |
| 201 | Bridgewire material improper | | | 3.2 | Certification |
| 202 | Evidence of poor workmanship | | 0.65% | 3.6 | Visual |

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# CLASSIFICATION OF DEFECTS & TESTS



| PARAGRAPH | TITLE | | SHEET 1 OF 1 | | DRAWING NUMBER |
|---|---|---|---|---|---|
| 4.4.2.3 | Bridged Header and Ignitor Assembly | | | | 9298600 |
| | | | | | NEXT HIGHER ASSEMBLY |
| | | | | | 9298551 |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| Critical | None defined | | | | |
| Major | | | | | |
| 101 | Lead styphnate on outside diameter of header | | 100% | 3.2 | Visual |
| 102 | Bridgewire visible through lead styphnate | | 0.40% | 3.2 | Visual |
| 103 | Lead styphnate cracked or separated (after drying) | | 0.65% | 3.2 | Visual |
| 104 | Moisture content of lead styphnate | | | 3.3 | 4.5.2 |
| Minor | | | | | |
| 201 | Evidence of poor workmanship | | 0.65% | 3.6 | Visual |

NOTES:





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**CLASSIFICATION OF DEFECTS & TESTS**

MIL-H-48384(AR)

| PARAGRAPH | TITLE | SHEET 1 OF 1 | DRAWING NUMBER |
|---|---|---|---|
| 4.4.2.4 | Header | | 0298602 |
| | | | NEXT HIGHER ASSEMBLY |
| | | | 0298600 |

| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
|---|---|---|---|---|---|
| Critical | None defined | | | | |
| Major | | | | | |
| 101 | Pull test of electrodes | 125 | 0.65% | 3.2 | 4.5.3 |
| Minor | | | | | |
| 201 | Evidence of poor workmanship | | 0.65% | 3.6 | Visual |

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QUALITY CONFORMANCE INSPECTION

# CLASSIFICATION OF DEFECTS & TESTS

MIL-H-48384(AR)

| PARAGRAPH | TITLE | | SHEET 1 OF 1 | | DRAWING NUMBER |
|---|---|---|---|---|---|
| 4.4.2.5 | Ignitor Assembly (prior to shorting) | | | | 9298551 |
| | | | | | NEXT HIGHER ASSEMBLY — |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| Critical | None defined | | | | |
| Major | | | | | |
| 101 | Bridgewire resistance | | 100% | 3.2 | 4.5.4 |
| 102 | Moisture content of lead styphnate | | | 3.3 | 4.5.2 |
| Minor | None defined | | | | |

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| PARAGRAPH | TITLE | | SHEET 1 OF 1 | | DRAWING NUMBER |
|---|---|---|---|---|---|
| 4.4.2.6 | Ignitor Assembly | | | | 9298557 |
| | | | | | NEXT HIGHER ASSEMBLY — |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| Critical | None defined | | | | |
| Major | | | | | |
| 101 | Assembly cracked, punctured or split | | 100% | 3.2 | Visual |
| 102 | Bare lead wires not shorted (twisted) together | | 100% | 3.2 | Visual |
| 103 | Cup crimp incomplete | | 0.40% | 3.2 | Visual |
| 104 | Epoxy seal missing or incomplete | | 0.40% | 3.2 | Visual |
| 105 | Insulation resistance | | 100% | 3.2 | 4.5.5 |
| 106 | Insulation resistance, thermal shock, vibration and function | 20 | 0.65% | 3.2 | 4.5.6 |
| 107 | All fire voltage (ambient) | 125 | .40% | 3.2 | 4.5.7 |
| 108 | Output pressure/time test | | | | |
| | Ambient (70° ± 5°F) | 80 | * | 3.2/3.4 | 28108397 or |
| | Low (-25° ± 5°F) | 20 | * | 3.2/3.4 | approved equivalent |
| 109 | Outside diameter, max, (for specified length) | | 0.40% | 3.2 | Gage |
| 110 | Length of closed cup, max. | | 0.40% | 3.2 | Gage |
| 111 | Length of insulation tubing | | 0.40% | 3.2 | Gage |
| Minor | | | | | |
| 201 | Evidence of poor workmanship | | 0.65% | 3.6 | Visual |

NOTES:
* All units must fire on first attempt. No failure-to-function allowed; one (1) pressure time failure permitted.



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## CLASSIFICATION OF DEFECTS & TESTS

MIL-H-48384(AR)

| PARAGRAPH | TITLE | | SHEET 1 OF 1 | | DRAWING NUMBER 9298550 |
|---|---|---|---|---|---|
| 4.4.2.7 | Cup, Sustainer | | | | NEXT HIGHER ASSEMBLY 9298549 |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| Critical | None defined | | | | |
| Major | | | | | |
| 101 | Material thickness | | 0.40% | 3.2 | Visual |
| 102 | Heat treatment | | 0.40% | 3.2 | Certification |
| Minor | | | | | |
| 201 | Evidence of Poor Workmanship | | 0.65% | 3.6 | Visual |

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## CLASSIFICATION OF DEFECTS & TESTS

MIL-H-48384(AR)

| PARAGRAPH | TITLE | SHEET | DRAWING NUMBER |
|---|---|---|---|
| 4.4.2.8 | Sustainer Charge Assembly (prior to packaging) | 1 OF 1 | 9298549 |
| | | | NEXT HIGHER ASSEMBLY: — |

| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
|---|---|---|---|---|---|
| Critical | None defined | | | | |
| Major | | | | | |
| 101 | Outside diameter | | 0.40% | 3.2 | Gage |
| 102 | Total length | | 0.40% | 3.2 | Gage |
| 103 | Charge above flush | | 0.40% | 3.2 | Visual |
| 104 | Assembly deformed or split | | 0.40% | 3.2 | Visual |
| 105 | Output pressure/time test | | | | |
| | Ambient (70°F ± 5°F) | 80 | * | 3.2/3.4 | 28108397 or |
| | Low (-25°F ± 5°F) | 20 | * | 3.2/3.4 | approved equivalent |
| 106 | Moisture content of black powder | | | | 4.5.1 |
| Minor | | | | | |
| 201 | Evidence of poor workmanship | | 0.65% | 3.6 | Visual |

NOTES:

* all units must fire on first attempt. No failure-to-function allowed; one (1) pressure time failure permitted.

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QUALITY CONFORMANCE INSPECTION

# CLASSIFICATION OF DEFECTS & TESTS

MIL-H-48384(AR)

| PARAGRAPH | TITLE | SHEET 1 OF 1 | DRAWING NUMBER |
|---|---|---|---|
| 4.4.2.9 | Carton (prior to sealing) | | See Section 5 |
| | | | NEXT HIGHER ASSEMBLY — |

| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
|---|---|---|---|---|---|
| Critical | None defined | | | | |
| Major | | | | | |
| 101 | Ignitor assemblies in bundle not completely encased in foil | | .40% | | Visual |
| 102 | Pocket assembly torn or insecurely closed (when applicable) | | .40% | | Visual |
| Minor | None defined | | | | |

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| PARAGRAPH | TITLE | SHEET 1 OF 1 | DRAWING NUMBER |
|---|---|---|---|
| 4.4.2.10 | Carton, sealed | | See Section 5 |
| | | | NEXT HIGHER ASSEMBLY — |

| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
|---|---|---|---|---|---|
| Critical | None defined | | | | |
| Major | | | | | |
| 101 | Carton damaged to extent that contents are exposed or liable to become exposed | | .40% | | Visual |
| Minor | | | | | |
| 201 | Marking missing, incorrect or illegible | | .65% | | Visual |

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| PARAGRAPH | TITLE | | SHEET 1 OF 1 | | DRAWING NUMBER |
|---|---|---|---|---|---|
| 4.4.2.11 | Sustainer Charge Assembly (after packing in container and prior to closing container) | | | | 9298549 |
| | | | | | NEXT HIGHER ASSEMBLY |
| | | | | | – |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| Critical | None defined | | | | |
| Major | | | | | |
| 101 | Assemblies not secure | | 0.40% | 5.2 | Visual |
| 102 | Dessicant missing | | 100% | 5.2 | Visual |
| Minor | None defined | | | | |

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4.4.3 Testing. - Testing is described in the First Article and Quality Conformance inspection Tables.

4.4.4 Inspection Equipment. - The inspection equipment required to perform the examinations and tests prescribed herein is described in the Paragraph Reference/Inspection Method column in the tables starting with Paragraph 4.4.2.1. The contractor shall submit for approval. inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Test methods and procedures. -

4.5.1 Moisture and volatiles content of black powder. - The contractor shall provide adequate controls to insure compliance with the applicable requirements. The moisture and volatiles content of the explosive charge shall be determined at the loading station at the time of loading. This determination shall be made at least once each four (4) hours of production. Sampling of the explosive charge shall be in accordance with MIL-STD-650, Method 101.5. In addition, if the moisture and volatiles content fails to comply with the requirements specified on the applicable drawing, and loading has not begun, the explosive charge material shall be rejected until corrective action has been taken as verified by repeating the moisture and volatiles content determination. If loading has begun, sustainer charge assemblies loaded with the non-conforming explosive charge shall be rejected.

4.5.2 Moisture content of lead styphnate. - The contractor shall provide adequate controls to insure compliance with the applicable requirements. The moisture content of both lead styphnates shall be determined at the loading station at the time of loading. This determination shall be made at least once each eight (8) hours of production. Sampling of both styphnates shall be in accordance with the applicable specifications. In addition, if the moisture content fails to comply with the requirement specified in 3.3, and loading has not begun, the styphnate shall be rejected until corrective action has been taken as verified by repeating the moisture content determination. If loading has begun, igniter assemblies loaded with the non-conforming styphnates shall be rejected. The moisture content shall be determined as follows: using appropriate safety precautions, weigh approximately two (2) grams of the plant-dried sample accurately within 0.1 mg into



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a tared stoppered weighing dish. The dish shall be covered with a sheet of polytetrafluoroethylene. The procedure of MIL-STD-650, Method 101.6 shall be used with the exception that the sample shall be dried at 60° ± 5°C, for three (3) hours.

4.5.3 Pull test of electrodes of header. -The header shall be assembled in an approved fixture and the procedure of MIL-STD-202, Method 211, Test Condition A shall be used with the exception that the applied force shall be 8.0 ± 0.1 pounds per electrode for 45 seconds. The force shall be a gradually applied static axial load. Any header in which either electrode moves or loosens shall be classed defective. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.5.4 Bridgewire resistance of igniter assembly. - Prior to shorting (twisting) the lead wires, the assembly shall be connected to approved equipment and the bridgwire (lead to lead) resistance measured to determine compliance with the requirement specified on the applicable drawing. The measurement shall be made with a meter capable of measurement to the nearest 0.01 ohm. The current applied shall be limited to one (1) milliampere, max.

4.5.5 Insulation resistance of Igniter Assembly. - The resistance between the shorted (twisted) lead wires and the igniter cup shall be determined using the procedure specified on the applicable drawing and MIL-STD-202, Method 302, Test Condition B. The specified voltage shall be applied for 2 seconds minimum. Examination shall be made to determine compliance with the insulation resistance requirement specified on the applicable drawing as well as any evidence of insulation failure such as loosening, cracking, or charring of the insulation tubing or reduced voltage.

4.5.6 Insulation resistance, thermal shock, vibration and function: The ignitor assemblies shall be tested as follows, in the sequence specified, using approved equipment:

a. Measure resistance between the shorted electrode and cup in accordance with 4.5.4.
b. Condition for two (2) hours minimum at -65 ± 5°F.
c. Remove from cold chamber and within 20 seconds transfer to a hot chamber at 165 ± 5°F.



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d. Condition for two (2) hours minimum at +165 ± 5°F and 20% maximum relative humidity.
e. Remove from hot chamber.
f. Vibrate for 15 minutes, minimum in a sweep covering a frequency range of 25-500-25 cycles per second, parallel. to the transverse orthogonal axis, with base charge down and a 2 ± 0.2 G's peak.
g. Measure resistance between electrode and cup in accordance with 4.5.5.
h. The igniters shall be functioned as specified in 4.5.7.

4.5.7 All-fire voltage of Igniter Assembly. - The ignitor assembly shall be subjected to an all-fire voltage of 1.6 volts maximum, by discharging a 100 microfarad capacitor impressed across the bridge circuit. Observation shall be made to determine compliance with the requirement specified on the applicable drawing. A record of each firing shall be kept.

4.5.8 Output pressure/time Test of Igniter Assembly and Sustainer Charge Assembly lots. - This test shall be performed for each combination of ignitor assembly and sustainer Charge assembly serial lots to be used in the Housing, Timing and Fuzing lot. The assemblies shall be assembled in a fixture in accordance with Dwg. 9298424 to simulate the volume of the sensor system and which is instrumented to provide a record of pressure versus time. The assemblies and fixture shall be conditioned at the required test temperature for two (2) hours minimum. Equipment, calibration and test procedures shall be documented and submitted in accordance with 4.4.4. The igniter assembly shall be initiated using a suitable power source and the pressure/time results evaluated against the applicable requirements. If any igniter-sustainer charge assembly fails to function and/or more than one (1) pressure/ time failure occurs, that combination of serial lots shall be rejected for use in the Housing, Timing, and Fuzing lot.

Care shall be taken during conditioning and handling of the assemblies to avoid contamination. The open end of the sustainer charge assembly shall at no time be contaminated with silicon lubricant. The fixture shall be cleaned after each shot or, as required, using the procedure specified in ES8673.

4.5.9 Check test for deterioration of igniter assemblies and sustainer charge assemblies. - The following test is designed to prevent igniter assemblies and sustainer charge



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assemblies that have deteriorated from being assembled into major ammunition items. The test shall be performed on igniter assemblies and sustainer charge assemblies from each lot which have been stored for more than two years, under adverse conditions for any period of time, or at the request of the Contracting Officer. If the specification for the ammunition item into which the igniter assemblies and sustainer charge assemblies are assembled does not require that the check test be performed, the contracting officer should insert in the contract a provision for the performance of the check test. The contracting officer will arrange for the contractor to be reimbursed for expense incurred in his performance of the check test for possible deterioration.

4.5.9.1 Functioning. - Seventy (70) igniter assemblies assembled with sustainer charge assemblies shall be selected from each lot and subjected to the functioning test specified in 4.5.8. If two (2) or more igniter assemblies fail to comply with the requirements specified on the applicable drawing, including duds, the lot shall be rejected. If any lot is rejected, all data generated shall be forwarded to Commander, ATTN: DRDAR-QAA-Q, ARRADCOM, Dover, N.J. 07801.

5. PACKAGING. -

5.1 Preservation and packaging. -

5.1.1 Level C

5.1.1.1 Sustainer charge assembly. - The sustainer charge assembly shall be preserved and packaged in accordance with Dwg. 9298549.

5.1.1.2 Igniter Assembly. -

(a) Rosett wrap. - A bundle of 25 igniter assemblies shall be made, lead wires out straight, using the lead wire from one igniter to secure the bundle. Each bundle of igniters shall be wrapped in a sheet of approximately 8 inches by 8 inches by .0015 inch thick aluminum foil complying with MIL-A-148, so that all igniters and lead wires in a bundle are completely encased by the foil. Five bundles of igniters, totaling 125 igniters, shall be packed in a folding carton, set-up box, or comparable container, made of material with a minimum board thickness of .03 inch.



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(b) Pocket assembly, alternate pack. - The ignitors shall be packed in pocket assemblies manufactured in accordance with Figure 4 (as applicable) made of 0015 inch aluminum foil complying with MIL-A-148. An adhesive seal shall be used to bond the faces of the aluminum foil together to form five pockets. The adhesive used for the sealing of the two faces of the aluminum foil shall be suitable for the application. The strength of the seal shall be sufficient to allow the insertion of the dowel (See Figure 4) without separation of the seal. The two faces of any seal shall not be misaligned by more than 1/16 inch when the pouch is formed. Each pocket shall be opened to form a cylinder using a smooth dowel with rounded end and similar to that shown on Figure 4. Each pocket may, however, be opened by other suitable methods which will result in providing a satisfactory pocket for insertion of the igniter without injury to the material. One igniter shall be inserted, igniter end down, in each pocket. The lip of the pocket assembly shall be folded over to form a closed pouch. Twenty-five pocket assemblies or a total of 125 igniters, shall be packed in a folding carton, set-up box, or comparable container, made of a material with a minimum board thickness of .03 inch.

(c) Marking. - Each inner carton, set-up box, or comparable container, shall be marked: "PATENTED -U.S. PATENT 2,389,086."

5.2 Packing. -

5.2.1 Level C. - The sustainer charge assembly and igniter assembly, preserved and packaged as specified in 5.1.1 shall be packed for shipment in accordance with Code of Federal Regulations, Title 49, Parts 100-199.

5.3 Marking. -

5.3.1 Igniter Assembly. - The cartons shall be marked in accordance with Dwg. 8796522. The DOT marking shall be: (QTY) - ELECTRIC BLASTING CAPS-HANDLE CAREFULLY

NOTE: Insert number of blasting caps in the exterior container.

5.3.2 Sustainer charge assembly. - The cartons shall be marked in accordance with Dwg. 9298549 and Code of Federal Regulations, Title 49, Parts 100-199.



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5.4 Shipping. - When components from more than one lot are shipped at one time, such as in a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Intended use. - The components covered by this specification are intended for use with Housing, Timing and Fuzing.

6.2 Ordering data. - See MIL-A-48078.

6.3 Submission of inspection equipment designs for approval. - See MIL-A-48078. Submit equipment designs as required to: Commander, ARRADCOM, ATTN: DRDAR-QAA-I, Dover, New Jersey 07801. Letter of submittal should state contractor, contract number, specification number, item nomenclature and classification of defects or test paragraph.

6,4 Drawings. - Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activites are now under the cognizance of ARRADCOM.

Custodian:
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Preparing Activity:
Army-AR

Project Number: 1345-A084



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NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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